

Video Transcript

Video 9: Redundant Dictionaries in Python Exercise (08:19)

So, let's see if we can make our dictionaries a little bit faster. Here we're going to use some redundant dictionaries. Now, we saw last time that we had dictionaries within dictionaries, that students itself is a dictionary. So, let's give ourselves a helper function here, where if we pass in the dictionary, we can pass two keys. For example, we might want to add a certain problem set to the problem sets. So, let's see how we do that. So, we're first going to check if key1 exists. So, if key1 not in the dictionary, then we put the key in there. We set up the dictionary.

Now, if it is in there, we go to it. And we add now to that dictionary a new key and we put it equal to the value that's being passed in. So, if the second key doesn't exist, it's going to create it and it's going to put the value in. So, this is the way that we initialize dictionaries.

So, that gives us a very nice helper function. Now, let's take a look at how we might add a student. We can borrow from the way we added student before, but let's just go through it. We're going to add to two dictionaries, studentById and studentByName. We're adding a student to two of those dictionaries. So, we're passing in a database. And we pass in the student_id and we pass in the student_name. Okay. Now what we want to do is we're going to go to the first dictionary, studentsById, and we're going to add the student_id and it's going to be equal to, this will be the student_name. And the next one we need db studentByName, and we're going to have student_name here, and that's going to be equal to student_id. So, now we've added a student. Let's now addPset. So, that's going to stay very much the same.

Okay. So, that's straightforward. Now, the last one, addGrade. Now, we're going to have gradesByStudent and gradesByPset. So, db student_id, pset_id and points. And now, we need to take advantage of our helper function. So, add2 db and now the first one is gradesByStudent. We're going to have student_id and points and this one is going to be add2 db again. But now, we're going to have gradesByPset and pset_id and we're missing one here. We need pset_id here. Because where this takes four arguments, key1 and key2. So key1, key2. And so, this one is just reversed. Now we've got pset_id, student_id points. Neat. Now, let's print the database. We haven't actually added the students, but let's see what we've got. We've got a typo. I've got a typo here somewhere there.

We're looking good. So, we've got an empty database. Now, we need to populate the database again. So, let's copy that from before. Up here we populated the database. Let's grab these again. Here's our redundant dictionaries. Let's go with all of these in here. I've got a typo here. I'm checking this. This is going to create the empty one. We should be fine. Let's take a look. Yes. Okay, now we've got `studentsById`, `studentsByName`, `gradesByStudent`, `gradesByPset`. Beautiful. Now we should be able to look them up much faster and much easier. Let's take a look at that. So, this is our database. So, let's write `getGradesOnPset`. So, we're going to pass in the database and we're going to pass in the `pset_id`.

Now we know we've already got here, we've got `gradesByPset` in the db database. So, we know that's what we want to use. But this will give us all the psets. Now, we want to pick out one key. So, remember we can use `get` on this. So, we pass in the dictionary and the key and the key is the `pset_id`. And of course we need to actually return it. So, now let's run this and you see we've got student one, student two, and student three. And if you check, that's exactly what we've got here, two and three. Okay, why don't you now write one that will print out instead of printing out by student ID, print out the names. So, what I want you to write is `getGradesOnPset` but where we construct and have this with a name instead of the one, two and three. Bye for now.